AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Strategic Datasphere, GA -- station KPDK, America/New_York
Committed pair	OSM 39083541 -> 615593145 Strategic Datasphere / Mapletree Industrial
Facade gap	137.2 m between the two halls
Weather record	43,250 real hourly records from KPDK
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.2723 C at 150 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2022-03-13 (crossing)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	1 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 5 of 23 hours with 1 mode change. The reactive incumbent took 15 hours -- 10 more -- but declared 3 unsafe hours against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 15 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	5 of 23 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	15 of 23 hours free cooling, 2 mode change(s), 3 unsafe hour(s) the incumbent broke its own switch budget 1 time(s) to stay safe 15 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	-3.232	18.0	-3.280	0.889
01	mechanical	yes	switch budget	-3.544	18.0	-3.890	0.761
03	mechanical	yes	switch budget	-3.514	18.0	-4.390	0.882
04	mechanical	yes	switch budget	-3.839	18.0	-5.000	0.816
05	mechanical	yes	switch budget	-4.117	18.0	-5.000	0.750
06	mechanical	yes	switch budget	-4.379	18.0	-5.000	0.660

07	mechanical	yes	switch budget	-4.449	18.0	-5.610	0.782
08	mechanical	yes	switch budget	-1.285	18.0	-2.780	1.501
09	mechanical	yes	switch budget	1.402	18.0	-0.610	2.200
10	mechanical	yes	switch budget	2.774	18.0	1.110	2.499
11	mechanical	yes	switch budget	4.213	18.0	3.344	1.650
12	mechanical	yes	switch budget	5.824	18.0	5.790	1.208
13	mechanical	yes	switch budget	14.211	18.0	21.932	1.055
14	mechanical	yes	switch budget	14.774	18.0	22.220	0.891
15	mechanical	yes	switch budget	15.476	18.0	21.683	1.047
16	mechanical	no	dry-bulb	21.399	18.0	18.560	1.012
17	mechanical	no	dry-bulb	23.376	18.0	23.042	1.118
18	mechanical	no	dry-bulb	22.271	18.0	22.780	1.157
19	FREE	yes	-	15.190	18.0	12.356	1.421
20	FREE	yes	-	15.195	18.0	7.916	1.619
21	FREE	yes	-	14.085	18.0	5.696	1.512
22	FREE	yes	-	7.970	18.0	3.466	1.435
23	FREE	yes	-	5.195	18.0	2.356	1.092

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -3.232 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -3.544 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -3.514 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -3.839 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -4.117 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -4.379 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -4.449 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -1.285 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 1.402 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.774 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.213 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

12:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.824 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

13:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.211 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

14:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.774 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

15:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.476 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.399 C against a 18.0 C limit, so it fails by 3.399 C. It would take a limit 3.399 C higher, or a bound 3.399 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.376 C against a 18.0 C limit, so it fails by 5.376 C. It would take a limit 5.376 C higher, or a bound 5.376 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.271 C against a 18.0 C limit, so it fails by 4.271 C. It would take a limit 4.271 C higher, or a bound 4.271 C tighter, to change this hour.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.190 C, which is 2.810 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.421 C, and the margin is measured, not chosen: 1.362 C of group-conditional forecast error for this hour of day, plus 0.0585 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.195 C, which is 2.805 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.619 C, and the margin is measured, not chosen: 1.560 C of group-conditional forecast error for this hour of day, plus 0.0585 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.085 C, which is 3.915 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.512 C, and the margin is measured, not chosen: 1.454 C of group-conditional forecast error for this hour of day, plus 0.0585 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.970 C, which is 10.030 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.435 C, and the margin is measured, not chosen: 1.376 C of group-conditional forecast error for this hour of day, plus 0.0585 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.195 C, which is 12.805 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.092 C, and the margin is measured, not chosen: 1.034 C of group-conditional forecast error for this hour of day, plus 0.0585 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

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